PredictIQ: Industrial IoT Predictive Maintenance Platform

Document: Architecture Technical Specification & Hackathon Jury Defense Guide
Author: Lead AI Systems Engineer | Target: Venture Innovation Jury

1. COMPLETE TECHNICAL STACK

A. Frontend (Digital Twin Console)

- Core Framework: React 19 with TypeScript in modern functional paradigm.
- Bundler & Dev Server: Vite 8.3 mounted via Express middleware on port 3000.
- Styling: Tailwind CSS v4 using an enterprise dark-slate palette (#0B0F17, #111722).
- Typography: IBM Plex Sans paired with JetBrains Mono for tabular metrics.
- Visualizations: Native responsive vector SVG coordinate-mapping system.

B. Backend Server & AI Copilot

- Server Runtime: Node.js with Express 4 REST architecture.
- AI SDK: @google/genai (v2.4.0) invoking gemini-3.8-flash for edge reasoning.
- Reliability: Fallback rule engine provides zero-latency diagnostic briefs.

C. Two-Stage Machine Learning Pipeline (PyTorch in app.py)

- Stage 1 (Health Index): Unsupervised Autoencoder (in:6 -> 16 -> 6 -> 16 -> out:6).
  Trained on early healthy cycles; reconstruction MSE acts as anomaly score.
  Statistical 3-sigma threshold triggers early knee-point degradation onset.
- Stage 2 (RUL Sequence Model): 2-layer GRU (in:6, hidden:32, layers:2).
  Operates on 30-cycle sliding windows; piecewise linear capping at 125 cycles.
- Optimization: Fast deterministic fitting (<2.5s) cached via Streamlit.

2. SUPPORTED MULTI-MACHINE DOMAINS

- Aviation Turbofans: T50 turbine temp, P30 compressor pressure, Nf fan speed, vibration.
- Power Transformers: Dissolved H2 gas ppm, moisture in oil, dielectric breakdown kV.
- Industrial Pumps: Inboard bearing temp, discharge flow m3/h, seal flush pressure.
- Wind Gearboxes: High-speed shaft temp, metal debris count, planetary vibration.

PredictIQ: Hackathon Jury Defense Guide (Part 1 - ML & Physics)

CATEGORY 1: MACHINE LEARNING & MODELING DEFENSE

Q1: Why use an Autoencoder for Stage 1 instead of standard One-Class SVM or Isolation Forest?

A: Industrial sensor degradation is multi-dimensional and continuous. Isolation Forests create rigid, step-function partition boundaries that fail to capture smooth physical wear. An Autoencoder models the non-linear correlations between sensors. Reconstruction MSE provides a smooth, physical Health Index that smoothly tracks clearance erosion and thermodynamic loss.

Q2: Why did you pick GRU/LSTM over Transformers for Remaining Useful Life (RUL)?

A: Industrial edge hardware (PLC / FADEC) has strict memory and latency bounds. For sequential window sizes of 30 cycles, self-attention mechanisms in Transformers introduce quadratic O(N^2) overhead and overfit on small fleets without providing measurable accuracy gains. GRUs converge in under 2.5 seconds on CPU with high inference speed (<2ms/window).

Q3: Why is RUL clipped at 125 cycles in the training pipeline?

A: In turbomachinery, brand-new components do not exhibit measurable wear during early cycles. Labeling an engine at cycle 5 with an RUL of 190 forces the neural network to learn an unobservable linear decay slope. Piecewise linear clipping (cap = 125) reflects the physical reality that wear only manifests after the operational baseline.

Q4: Explain the NASA C-MAPSS Asymmetric Scoring metric and why MSE is dangerous in aerospace.

A: Symmetric MSE treats predicting 10 cycles too late identically to 10 cycles early. In aviation, predicting late (d > 0) means an engine fails in-flight before maintenance is scheduled! NASA penalty penalizes late calls exponentially (s = exp(d/13)-1) much harsher than early calls (s = exp(-d/10)-1).

PredictIQ: Hackathon Jury Defense Guide (Part 2 - Systems & ROI)

CATEGORY 2: FULL-STACK & EDGE SYSTEMS ARCHITECTURE

Q5: How does the system prevent LLM hallucinations in the Machine Copilot?

A: The Copilot does not invent data. All telemetry (current readings, MSE, 3-sigma limits, RUL) is calculated deterministically by our mathematical models and injected as structured JSON context into the server-side Gemini prompt. The model acts strictly as an analytical explainability layer.

Q6: How do you handle transient sensor spikes (noise vs true failure onset)?

A: Individual sensor readings are subject to electrical EMI noise. Our Autoencoder reconstruction evaluates cross-sensor correlation across all 6 channels simultaneously. A single noisy pin will not breach the statistical 3-sigma multi-channel barrier without correlated shifts in related channels.

CATEGORY 3: BUSINESS VALUE & COMMERCIAL ROI

Q7: How do you justify the $168,000 cost avoidance figure to a CFO?

A: An unscheduled Aircraft on Ground (AOG) or transformer flashover averages $180,000 to $450,000 in emergency logistics, passenger rerouting, collateral core destruction, and grid fines. A scheduled overnight line replacement costs $12,000 to $22,000. Preventing just one catastrophic failure pays for our entire enterprise monitoring deployment for over 3 years (+1,400% ROI).

Q8: How does the platform scale from aviation to other machines like transformers or pumps?

A: Our two-stage architecture is domain-agnostic. The Autoencoder does not know what a turbine is; it learns normal correlation manifolds. Whether inputs are temperatures, dissolved hydrogen gas, or pump flow rates, the mathematical pipeline operates identically.